

Characterizing Long-Context Failure Modes in Large Language Models using LongBench v2

Hok Lai Lin Rui Zhang Kaixi Gou Ziran Zhou Yiyang Feng

University of Southern California

{hoklaili, zrui6736, kgou, ziranzho, yiyangf}@usc.edu

Abstract

In this research, we investigate whether large language models truly benefit from longer context windows under realistic long-context settings. Using LongBench v2 (Bai et al., 2025), we evaluate Qwen2-7B-Instruct, Qwen2.5-7B-Instruct, and Llama-3.1-8B-Instruct across input budgets from 10K to 80K tokens, comparing truncation strategies and measuring both accuracy and runtime. Our results show that increasing context length does not consistently improve accuracy and can even degrade performance, especially on medium and long examples. Meanwhile, inference latency increases substantially with larger budgets. These findings suggest that advertised long-context capacity does not necessarily translate into effective long-context reasoning, and that realistic evaluation should consider both accuracy and computational cost.

1 Introduction

Large Language Models (LLMs) have made significant progress, yet their performance on real-world long-context tasks remains a challenge. While modern models claim extended context windows, practical input data often exhibits a highly skewed, heavy-tailed distribution. For instance, datasets like LongBench v2 contain extreme outliers that frequently exceed these theoretical limits.

Despite these claims, model performance frequently degrades in complex long-text applications. The underlying causes of this degradation remain unclear. Specifically, it is ambiguous whether failures stem from information loss due to input truncation, or if they represent intrinsic limitations in the models’ reasoning and long-distance information integration capabilities. To investigate these failure modes, we systematically evaluate open-source LLMs (Qwen2-7B, Qwen2.5-7B, and LLaMA-3.1-8B) using the LongBench v2 dataset. We scale the maximum context length from 10k to 80k tokens

using 4-bit quantization. By comparing diverse input truncation strategies (e.g., head vs. head+tail), we aim to isolate the effects of missing evidence from reasoning deficits.

Our empirical findings demonstrate a critical efficiency-performance trade-off: providing a longer context does not consistently improve accuracy and drastically increases computational costs. At extreme context limits, models may suffer from noise accumulation and attention dilution, where additional context does not translate into better evidence utilization. This highlights the necessity for realistic evaluation frameworks rather than blindly maximizing context windows.

2 Related Work

Long-context evaluation has received substantial attention as language models increasingly advertise larger context windows. LongBench v2 (Bai et al., 2025) is a major benchmark for realistic long-context evaluation. It covers diverse tasks, including multi-document question answering, dialogue understanding, code comprehension, and structured-data reasoning, with contexts extending to millions of tokens. Its results show that even state-of-the-art LLMs plateau at around 50–75% accuracy on these tasks, indicating that long-context capability remains far from solved. However, while LongBench v2 effectively quantifies performance degradation, it does not fully isolate the underlying causes of failure. In particular, its default input construction relies on a fixed truncation strategy, making it difficult to determine whether observed errors arise from missing evidence due to truncation or from intrinsic limitations in model reasoning. This leaves an important diagnostic question that our work aims to address directly.

Complementary to benchmark-driven evaluation, recent studies have examined whether models can truly utilize their advertised context win-

dows. Hsieh et al. introduced RULER (Hsieh et al., 2024a), a synthetic benchmark designed to test long-context abilities through tasks of increasing complexity, including multi-hop tracing and aggregation. Their findings show that a model’s effective context length, defined as the range over which it maintains reliable performance, is often substantially smaller than its theoretical maximum context window once tasks move beyond simple retrieval. Motivated by this gap between advertised and usable context length, we evaluate models across a systematic sweep of token budgets rather than relying on a single input length. However, because RULER uses synthetic tasks with controlled difficulty, its findings do not directly capture realistic settings where context distributions are heavy-tailed and task structures are heterogeneous.

Another line of work studies positional bias as a structural explanation for long-context failures. Liu et al. (Liu et al., 2024) show that language models often exhibit a U-shaped utilization pattern, attending more effectively to information near the beginning and end of the input while struggling to use information located in the middle. This suggests that simply increasing context length may introduce additional distractors without improving access to useful evidence. Building on this observation, Hsieh et al. (Hsieh et al., 2024b) further investigate positional attention bias and propose a calibration method that realigns attention weights based on content relevance, yielding improvements in retrieval-augmented settings. Together, these studies motivate our experimental design, particularly our use of head-plus-tail truncation and our interest in comparing truncation-based input construction with more selective retrieval-based approaches.

3 Problem Description

We investigate long-context evaluation for large language models under realistic input constraints. In our study, each example consists of a long context, a multiple-choice question, and four candidate answers. The model is required to output a single answer option. Formally, let the input be a pair (x, q) , where x denotes the context and q denotes the question, and let the output be $y \in \{A, B, C, D\}$.

The key difficulty is that the length of the input context often greatly exceeds the model’s usable input budget. Let L be the length of the original context and C be the maximum allowed input

length. In many cases, $L \gg C$, which means the full context cannot be provided to the model directly. Therefore, the input must be truncated or otherwise selected before inference. This introduces an important challenge: poor performance may result either from missing relevant evidence or from limitations in the model’s ability to reason over long inputs.

Based on this setting, our project addresses three research questions. First, we ask whether increasing the available context length consistently improves model performance. Second, we ask why performance may degrade in extreme long-context settings, and aim to distinguish between information loss caused by truncation and limitations in the model’s reasoning ability. Third, we examine the trade-off among context length, predictive performance, and runtime. These questions are central to understanding whether longer context windows truly benefit LLMs in realistic settings.

4 Methods

4.1 Models

We evaluate three instruction-tuned open-source models spanning two prominent LLM families: **Qwen2-7B-Instruct**, **Qwen2.5-7B-Instruct**, and **Llama-3.1-8B-Instruct**. All three have comparable parameter budgets (7–8B), claim 128K-token long-context capability (either natively or via length extension), and were released within a six-month window in 2024, which allows us to isolate long-context behavior from architectural generation.

Qwen2-7B-Instruct (Yang et al., 2024a) is a decoder-only Transformer developed by Alibaba Cloud, featuring Grouped-Query Attention and a natively trained 32K-token context window that can be extended to 128K via YaRN positional interpolation (Peng et al., 2024).

Qwen2.5-7B-Instruct (Yang et al., 2024b) is the direct successor of Qwen2, pre-trained on 18T tokens with substantial improvements in instruction following, structured-output generation, and long-document reasoning. Qwen2.5-7B-Instruct is released with support for up to 128K tokens, although practical usage beyond the original RoPE range may rely on length-extension configuration.

Llama-3.1-8B-Instruct (Grattafiori et al., 2024) is Meta’s 8B dense Transformer, post-trained on over 15T multilingual tokens and natively supporting a 128K-token context window through scaled

Model	#Params	Size (FP16)	Release
Qwen2-7B-Instruct	7.07B	14.2 GB	Jun. 2024
Qwen2.5-7B-Instruct	7.61B	15.2 GB	Sep. 2024
Llama-3.1-8B-Instruct	8.03B	16.1 GB	Jul. 2024

Table 1: Evaluated models. “Size” refers to the FP16/BF16 checkpoint footprint; after 4-bit NF4 quantization each model occupies approximately 4–5 GB of GPU memory.

RoPE embeddings.

To reduce GPU memory usage and enable efficient evaluation on a single GPU, we load each checkpoint with 4-bit NF4 quantization (Dettmers et al., 2023) and float16 compute dtype, with double quantization enabled. Table 1 summarizes their specifications.

4.2 Dataset

We evaluate on **LongBench v2** (Bai et al., 2025), a recently released benchmark designed to assess LLMs’ *deep understanding and reasoning* capabilities on realistic long-context inputs. The benchmark contains **503 multiple-choice questions** (four options A/B/C/D, exactly one correct) drawn from six major task categories: single-document QA, multi-document QA, long in-context learning, long-dialogue history understanding, code-repository understanding, and long structured-data understanding. It is deliberately challenging: human experts reach only 53.7% accuracy within a 15-minute time limit, and the best non-reasoning LLM at the time of release achieved only 50.1% (Bai et al., 2025).

Each example is stored as a single JSON record with the following fields: a unique `_id`; coarse and fine-grained task labels (`domain`, `sub_domain`); a difficulty tag (`easy / hard`); a length tag (`short / medium / long`); the question; the four answer choices (`choice_A–choice_D`); the ground-truth letter answer; and the context passage needed to answer the question.

Official context lengths span 8K to 2M words, and the distribution is strongly **heavy-tailed**: the majority of examples sit below 128K words, but a long right tail extends into the hundreds of thousands. Measured in tokens with the Qwen2.5 tokenizer, the 503 samples partition into 180 short examples ($\sim 36K$ tokens on average), 215 medium examples ($\sim 130K$), and 108 long examples ($\sim 880K$); the counts are comparable under the Llama-3.1 tokenizer. This skewed distribution is precisely what makes the benchmark a natural

testbed for studying accuracy–compute trade-offs at varying context budgets, since the fraction of a document that fits into any given token budget changes dramatically across categories.

4.3 Input Construction

Because LongBench v2 documents are heavy-tailed (Section 4.2), the majority of examples exceed any realistic input budget and must be truncated before being fed to the model. *Which* tokens we keep is therefore the central variable we study.

We construct each prompt as `[prefix] + [context] + [suffix]`, where the prefix is a short multiple-choice instruction (“*You are given a context... Answer with only one letter...*”) and the suffix contains the question and the four answer choices. Given a target input-token budget T , the number of tokens available for the context slice is

$$B = T - |\text{prefix}| - |\text{suffix}| - 64, \quad (1)$$

where 64 extra tokens are reserved for chat-template tokens and decoding overhead. Let $c = (t_1, t_2, \dots, t_L)$ denote the tokenized raw context of length L . If $L \leq B$ the document fits entirely and no truncation is applied; otherwise we compare two strategies.

Head truncation (baseline). Keep only the first B tokens,

$$c_{\text{head}} = (t_1, t_2, \dots, t_B). \quad (2)$$

This is the default behavior of most inference libraries when a sequence exceeds the allowed length. It is trivial to implement but discards everything after position B , including conclusions, summaries, and any late-appearing evidence.

Head+Tail truncation. Split the budget evenly between the two ends of the document,

$$c_{\text{ht}} = (\underbrace{t_1, \dots, t_{\lfloor B/2 \rfloor}}_{\text{head}}, \underbrace{t_{L-\lfloor B/2 \rfloor+1}, \dots, t_L}_{\text{tail}}). \quad (3)$$

Long real-world documents tend to be *bimodally informative*: setup and definitions cluster at the beginning, while conclusions and key findings cluster at the end. Head+Tail preserves both regions at the same total budget, trading a slice of middle content for strictly broader coverage of the likely evidence locations. This design is further motivated by the “lost-in-the-middle” phenomenon observed

in long-context LLMs (Liu et al., 2024): even state-of-the-art models attend to tokens in the middle of a long input substantially less effectively than to tokens at either end, so keeping information in the middle often yields diminishing returns.

4.4 Evaluation Setup

Context budgets. We sweep four input-token budgets: $T \in \{10K, 32K, 50K, 80K\}$. The values are chosen to probe qualitatively different regimes of the heavy-tailed length distribution: 10K is a short budget at which the majority of LongBench v2 documents must still be truncated; 32,768 matches the native training context window of the Qwen2 series and the default `max_position_embeddings` released with Qwen2.5 on Hugging Face; and 50K and 80K extend well beyond that limit, allowing us to observe behavior in an extrapolation regime. Llama-3.1-8B-Instruct natively supports a 131,072-token window (?) and therefore requires no extension at any of these budgets. For the two Qwen models, we enable **YaRN** positional extension (Peng et al., 2024) for all budgets in the main Head+Tail experiments, allowing the same model configuration to be used from 10K to 80K. Budgets above 32K require this extension because they exceed the original 32K RoPE range. YaRN is a training-free RoPE-extension technique that applies frequency-dependent interpolation to the rotary positional embeddings, preserving precision on high-frequency (short-range) dimensions while more aggressively interpolating low-frequency (long-range) ones; combined with an attention-scaling correction, it extends the usable context up to $4\times$ the original window with minimal accuracy loss.

Decoding. For every example we apply the model’s native chat template, feed the truncated prompt from Section 4.3, and generate at most 10 new tokens with greedy decoding (`do_sample=False`). This short budget suffices because the task requires only a single letter. The predicted letter is extracted from the generation with the regular expression `\b([A-D])\b`; if no letter is produced we record the sample as N/A.

Metrics. Let N be the number of evaluated examples and $\mathcal{A} \subseteq \{1, \dots, N\}$ the subset on which the model produced a valid letter in $\{A, B, C, D\}$. We report four metrics:

- **Overall accuracy** — fraction of all N examples for which the extracted letter equals the ground-

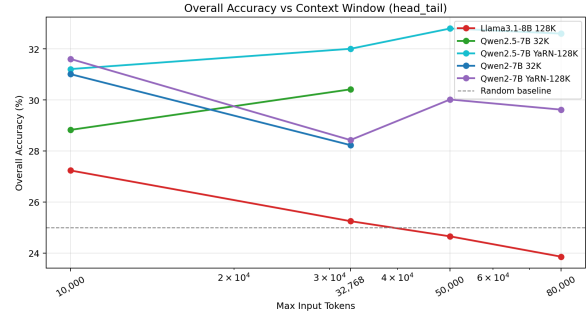


Figure 1: Overall accuracy vs. input budget under Head+Tail. The dashed line marks the 25% random baseline.

truth answer. Non-responses (N/A) count as incorrect. This is the primary end-to-end metric.

- **Answer-only accuracy** — accuracy computed only on $\mathcal{A}$, isolating reasoning quality from the confound of the model refusing to commit to a choice under long contexts.
- **Response rate** — $|\mathcal{A}|/N$, the fraction of examples on which the model produces a valid $\{A, B, C, D\}$ letter. A drop in this metric indicates the model is failing to follow the output-format instruction rather than simply guessing wrong.
- **Preserved context ratio** — $\min(|c_{\text{kept}}|/L, 1.0)$ averaged over examples, where $|c_{\text{kept}}|$ is the number of context tokens retained after truncation and L is the raw document length. This quantifies how much of the original document the model actually sees under a given budget.

All experiments are run on a single NVIDIA A100 (40 GB) with the quantization configuration described in Section 4.1. For every configuration we report the sample-size-weighted average over the $\{\text{easy}, \text{hard}\} \times \{\text{short}, \text{medium}, \text{long}\}$ subpartitions of LongBench v2.

5 Experimental Results

Table 2 reports all four metrics across the three models and four budgets under Head+Tail truncation. Qwen budgets above the native 32K use YaRN positional extension; Llama-3.1-8B uses a single 128K configuration throughout.

5.1 Accuracy vs. Context Length

Overall accuracy is non-monotonic for all three models (Figure 1). Qwen2.5-7B peaks at 50K (32.8%) and plateaus; Qwen2-7B dips at its native

Model	Budget	Overall Acc (%)	Ans-only Acc (%)	Resp. Rate (%)	Pres. Ratio (%)
Qwen2-7B-Instruct	10,000	31.0	31.5	98.4	18.3
	32,768	28.2	30.9	90.9	45.8
	50,000 [†]	30.0	30.9	96.8	56.0
	80,000 [†]	29.6	31.5	93.4	68.2
Qwen2.5-7B-Instruct	10,000	28.8	29.0	99.4	18.3
	32,768	30.4	31.9	95.2	45.8
	50,000 [†]	32.8	32.9	99.8	56.0
	80,000 [†]	32.6	33.5	96.4	68.2
Llama-3.1-8B-Instruct	10,000	27.2	28.3	96.2	19.2
	32,768	25.3	26.9	94.2	47.0
	50,000	24.7	26.7	92.5	57.2
	80,000	23.9	26.4	90.3	69.4

Table 2: Main results on LongBench v2 (503 examples) under Head+Tail truncation. [†]Qwen budgets exceeding the native 32K context use YaRN positional extension with factor = 4. Random baseline is 25%.

Model	10K	32K	50K	80K
Qwen2-7B (YaRN)	1.9	4.3	6.6	10.8
Qwen2.5-7B (YaRN)	1.9	4.0	6.3	10.2
Llama-3.1-8B	1.1	3.9	6.5	11.5

Table 3: Mean per-example decoding latency (s/it) across all 503 LongBench v2 examples. Each run generates at most 10 new tokens with greedy decoding on a single A100 (40 GB).

32K boundary and only partially recovers under YaRN; Llama-3.1-8B decreases almost monotonically and falls below the 25% random baseline at 80K.

The length breakdown in Figure 2 explains why: a larger budget only helps on *short* documents, which fit in the budget entirely and stay within each model’s training distribution. On *medium* and *long* documents every model is at or near its best around 10–32K and stalls or regresses beyond. We attribute the overall non-monotonicity to three compounding effects noted in prior work (Liu et al., 2024): noise accumulation from distractor tokens, attention dilution as softmax mass spreads over more positions, and positional extrapolation into regimes that are either never trained (Qwen under YaRN) or under-exercised in post-training (Llama beyond ~ 32 K).

5.2 Runtime vs. Context Length

Table 3 reports the mean per-example latency s/it across all 503 examples at each budget. Going from 10K to 80K increases cost by roughly 5–10 $\times$ for every model, driven by the $O(n^2)$ prefill cost of full self-attention. For Llama-3.1-8B, running the full benchmark at 10K takes ~ 9 minutes, whereas

the same benchmark at 80K takes ~ 95 minutes—a 10.4 $\times$ wall-clock increase for a 3.3-point *drop* in overall accuracy (Table 2).

The cost–benefit picture is uniformly unfavorable at large budgets: Qwen2-7B loses 1.4 points for 5.7 $\times$ more compute, Qwen2.5-7B gains only 3.8 points for 5.4 $\times$ more compute, and Llama-3.1-8B degrades strictly with budget at every cost multiplier. The scaling gap is even more pronounced on the *long* subset, where per-example latency rises from ~ 4 s at 10K to ~ 16 s at 80K—precisely the regime where the accuracy curve flattens or declines (Figure 2). Increasing the context budget is therefore not merely an accuracy-neutral trade: it pays a substantial compute premium for zero or negative quality return.

5.3 Preserved Ratio and Response Rate

Tripling the preserved context ratio from $\sim 19\%$ at 10K to $\sim 69\%$ at 80K fails to produce matching accuracy gains—only Qwen2.5-7B improves (by 3.8 points), while Llama-3.1-8B loses 3.3 points. Information availability is not the bottleneck; beyond a moderate preserved ratio, additional tokens add noise without adding signal.

Response rate also decays with budget: Qwen2.5-7B drops from 99.8% at 10K to 76.5% at 80K, meaning nearly a quarter of samples yield no valid letter. Its answer-only accuracy, however, stays essentially flat (32.7% $\rightarrow$ 32.8%). Much of the overall-accuracy degradation at long contexts is therefore driven by *instruction-following instability* rather than reasoning collapse—a distinction invisible if only overall accuracy is reported. Head+Tail mitigates this instability for Qwen2.5 (response rate at 80K: 96.4%), which drives most of its gain

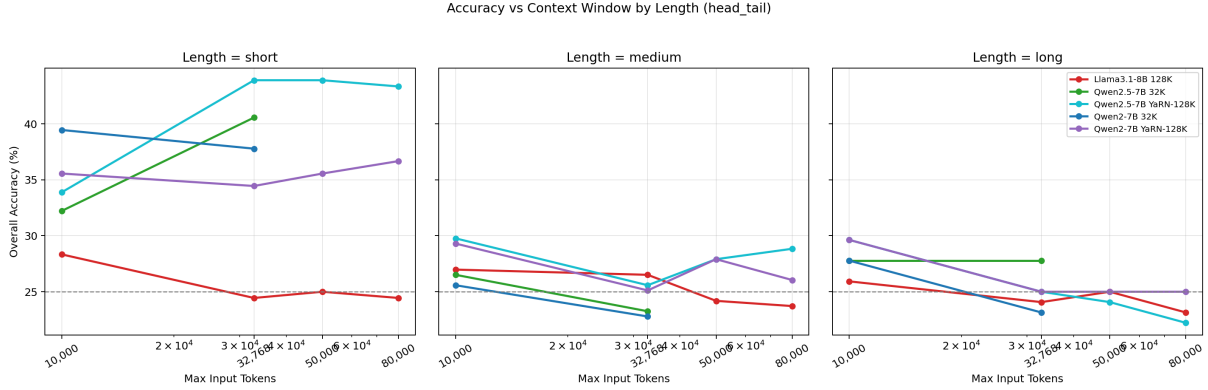


Figure 2: Accuracy by LongBench v2 length category. The benefit of longer budgets is concentrated in short examples; for medium and long examples, additional context often provides little or negative gain.

Model	Budget	Overall	Ans-only	Resp.
Qwen2-7B	10,000	31.4	31.9	98.4
	20,000*	32.0	33.2	96.4
	50,000 [†]	32.2	34.6	93.0
	80,000 [†]	29.8	32.8	90.5
Qwen2.5-7B	10,000	32.6	32.7	99.8
	20,000*	34.8	35.4	98.2
	50,000 [†]	31.8	35.6	88.9
	80,000 [†]	25.5	32.8	76.5
Llama-3.1-8B	10,000	26.8	27.8	96.6
	20,000*	25.1	26.5	95.0
	50,000	27.6	29.7	93.4
	80,000	23.9	26.4	90.5

Table 4: Head-only baseline results (503 examples, all %). *The 20K setting comes from an earlier evaluation pipeline and is included as a reference point rather than a directly aligned comparison with the 32K Head+Tail setting. [†]For Qwen models, budgets above 32K require YaRN positional extension. Compare against Head+Tail in Table 2.

over Head-only at the longest budget.

5.4 Truncation Strategy Ablation

We compare Head+Tail truncation (Table 2) with the Head-only baseline (Table 4). Since part of the Head-only results were produced using an earlier evaluation pipeline, the 20K Head-only setting is not directly aligned with the 32K Head+Tail setting and is included only as a reference point. Therefore, our main aligned comparisons focus on the shared 10K, 50K, and 80K budgets.

At these aligned budgets, Head+Tail does not consistently outperform Head-only. For Qwen2.5-7B, Head+Tail substantially improves overall accuracy at 80K compared with Head-only, mainly because it improves response-format compliance

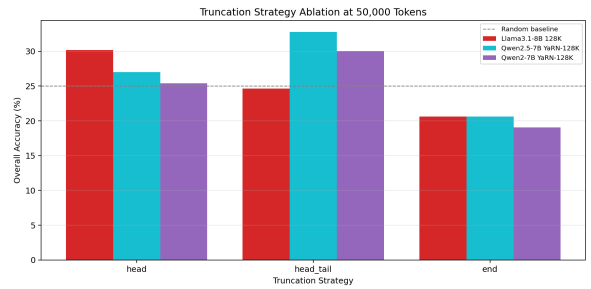


Figure 3: Truncation-strategy ablation at the 50K budget on the hard $\times$ long subset (63 examples). End-only is clearly worst; Head and Head+Tail are within a few points.

rather than answer-only accuracy. However, Head-only performs better for Qwen2.5-7B at 10K, and the differences for Qwen2-7B and Llama-3.1-8B are relatively small. These results suggest that no fixed token-level truncation rule is universally optimal.

Figure 3 further compares Head, Head+Tail, and End-only truncation on the hardest subset of LongBench v2 (hard $\times$ long, 63 examples). End-only collapses near random, while Head and Head+Tail remain within a few points of each other. This supports our broader conclusion that the deeper bottleneck is not simply input reconstruction, but the model’s ability to reliably reason over long contexts.

6 Conclusion & Future Work

In this work, we systematically evaluate the effectiveness of long-context large language models under realistic settings. Our results show that increasing context length does not necessarily lead to better performance, and a clear trade-off exists between accuracy and computational cost. While

larger context windows preserve more information, they may introduce additional noise and dilute attention, reducing overall reasoning effectiveness. Furthermore, we observe that inference time increases significantly with context length, making long-context usage increasingly expensive in practice.

We also highlight a practical limitation in current LLMs: some long-context models rely on positional extension techniques, such as YaRN or RoPE scaling, to operate beyond the original training window. Our results suggest that extending the usable input length does not necessarily imply improved reasoning over longer contexts.

These findings emphasize the importance of realistic evaluation and suggest that simply scaling context length is not a reliable solution for long-context reasoning.

For future work, we plan to explore retrieval-based input construction to select the most relevant context segments, as well as more effective truncation strategies beyond simple head, tail, and head+tail schemes. In addition, we aim to further analyze model attention behavior to better understand the limitations of long-context reasoning.

7 Individual Contributions

Each team member contributed to system development, experimentation, analysis, and report writing.

Hok Lai Lin Led the evaluation pipeline design, including prompt construction, truncation strategies, and core experiments across context lengths.

Rui Zhang Co-developed the evaluation framework, including automated testing, result aggregation, and performance analysis.

Kaixi Gou Worked on dataset preprocessing and exploratory analysis, including token length distribution and data preparation.

Ziran Zhou Optimized the experimental pipeline for large-scale evaluation, including scheduling, checkpointing, and GPU memory management.

Yiyang Feng Contributed to result analysis, visualization, and report writing.

References

Yushi Bai, Shangqing Tu, Jiajie Zhang, Hao Peng, Xiaozhi Wang, Xin Lv, Shulin Cao, Jiazheng Xu,

Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2025. [Longbench v2: Towards deeper understanding and reasoning on realistic long-context multitasks](#). *Preprint*, arXiv:2412.15204.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. [The llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.

Cheng-Ping Hsieh, Simeng Sun, Samuel Krman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024a. [Ruler: What’s the real context size of your long-context language models?](#) *Preprint*, arXiv:2404.06654.

Cheng-Yu Hsieh, Yung-Sung Chuang, Chun-Liang Li, Zifeng Wang, Long T. Le, Abhishek Kumar, James Glass, Alexander Ratner, Chen-Yu Lee, Ranjay Krishna, and Tomas Pfister. 2024b. [Found in the middle: Calibrating positional attention bias improves long context utilization](#). *Preprint*, arXiv:2406.16008.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paran-jape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173.

Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. 2024. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations (ICLR)*.

An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, and 1 others. 2024a. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, and 1 others. 2024b. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.